MA1220 Final Exam Autumn 2026: Answer Key and Marking Scheme

Oct 4, 2026

Model answers and mark splits for all 8 questions, 40 marks in total.

Q1. Definitions (5 marks)

Each part carries 1 mark: full marks for a complete, correct definition; half a mark if one essential condition is missing.

Part	Model answer	Marks	Half-mark deduction if
(a)	For a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$, let m_i and M_i be the infimum and supremum of f on $[x_{i-1}, x_i]$. Then $L(f,P) = \sum m_i(x_i - x_{i-1})$ and $U(f,P) = \sum M_i(x_i - x_{i-1})$. f is Riemann integrable if $\sup_P L(f,P) = \inf_P U(f,P)$; the common value is $\int_a^b f$. (The limit of Riemann sums as the mesh tends to 0 is also accepted.)	1	Only the sums are written, with no equality of sup and inf
(b)	$\int_a^\infty f$ is defined as $\lim_{t\to\infty} \int_a^t f$. If this limit exists and is finite the integral converges, otherwise it diverges.	1	Limit stated without saying it must be finite
(c)	f is continuous at (a,b) if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\lvert f(x,y) - f(a,b)\rvert < \varepsilon$ whenever $\sqrt{(x-a)^2 + (y-b)^2} < \delta$.	1	Quantifiers in the wrong order or distance condition missing
(d)	For a unit vector $u = (u_1, u_2)$, the directional derivative is $D_u f(a,b) = \lim_{h\to 0}\big[f(a+hu_1, b+hu_2) - f(a,b)\big]/h$, provided the limit exists.	1	u not stated to be a unit vector
(e)	f has a local maximum at (a,b) if there is a disc around (a,b) such that $f(x,y) \le f(a,b)$ for all (x,y) in the disc (within the domain). A local minimum is the same with $f(x,y) \ge f(a,b)$.	1	Only one of the two is defined, or the neighbourhood is missing

Q2. Riemann sums ($2 + 4 = 6$ marks)

(a) The integral of x^3 from 0 to 2 equals 4.

$$\int_0^2 x^3\,\mathrm{d}x = \lim_{n\to\infty}\sum_{i=1}^{n}\left(\frac{2i}{n}\right)^3\frac{2}{n} = \lim_{n\to\infty}\frac{16}{n^4}\left[\frac{n(n+1)}{2}\right]^2 = \lim_{n\to\infty}4\left(\frac{n+1}{n}\right)^2 = 4$$

Step	Marks
$\Delta x = 2/n$, right endpoints $x_i = 2i/n$, and the sum written as $(16/n^4)$ times the sum of i^3	1
Uses the sum of cubes formula and evaluates the limit to 4	1

(b) g is the constant equal to the integral of $\ln f(t)$ from 0 to 1.

$$\ln L_n = \frac{1}{n}\sum_{k=1}^{n}\ln f\left(\frac{k}{n}\right) \longrightarrow \int_0^1 \ln f(t)\,\mathrm{d}t, \qquad \text{so} \quad \lim_{n\to\infty} L_n = \exp\left(\int_0^1 \ln f(t)\,\mathrm{d}t\right)$$

Step	Marks
Takes logarithms of the product to get $(1/n)$ times the sum of $\ln f(k/n)$	1
Notes $f > 0$ and continuous, so $\ln f$ is continuous on $[0,1]$ and the sum is a Riemann sum	1
Concludes the sum tends to the integral of $\ln f$ over $[0,1]$	1
Uses continuity of exp to get $g = \int_0^1 \ln f(t)\,\mathrm{d}t$	1

The right side of the limit does not depend on x, so full marks go to the constant answer. Do not penalise a student who writes $g(x) =$ that integral, but a student who searches for a non-constant $g(x)$ should lose the last mark.

Q3. Arc length and area ($3 + 3 = 6$ marks)

(a) The length is $3/2 + (1/4)\ln 2$.

$$y' = x - \frac{1}{4x}, \qquad 1 + (y')^2 = \left(x + \frac{1}{4x}\right)^2, \qquad L = \int_1^2 \left(x + \frac{1}{4x}\right)dx = \left[\frac{x^2}{2} + \frac{1}{4}\ln x\right]_1^2 = \frac{3}{2} + \frac{1}{4}\ln 2$$

Step	Marks
Correct derivative $y' = x - 1/(4x)$ and the formula $L = \int \sqrt{1 + y'^2}$	1
Shows $1 + y'^2$ is a perfect square and takes the root as $x + 1/(4x)$, noting $x > 0$	1
Integrates and evaluates to $3/2 + (1/4)\ln 2$	1

Sign of the answer: if the final value has a sign change (for example $3/2 - (1/4)\ln 2$ instead of $3/2 + (1/4)\ln 2$), full marks are not given. Reduce the last mark by 1/2.

(b) The area is 18 square units.

$$y^2 - 2y - 8 = 0 \Rightarrow y = -2,\ 4; \qquad A = \int_{-2}^{4} \left[(y+1) - \frac{y^2-6}{2}\right] dy = \int_{-2}^{4} \left(-\frac{y^2}{2} + y + 4\right) dy = 18$$

Step	Marks
Finds the intersection points $y = -2$ and $y = 4$, that is $(-1, -2)$ and $(5, 4)$	1
Sets up the area integral with the line on the right of the parabola (integrating in y)	1
Evaluates the integral to 18	1

Integrating in x is also accepted. It needs two pieces: from $x = -3$ to -1 the region lies between $y = -\sqrt{2x+6}$ and $y = \sqrt{2x+6}$, and from $x = -1$ to 5 between $y = x - 1$ and $y = \sqrt{2x+6}$. Give the setup mark only if both pieces are correct.

Sign of the answer: if the final value has a sign change (for example -18 instead of 18, from taking the curves in the wrong order), full marks are not given. Reduce the last mark by 1/2.

Q4. Improper integrals ($3 + 3 = 6$ marks)

(a) The integral of $\sin x \cos x/x$ from 1 to ∞ converges.

$$\int_1^\infty \frac{\sin x \cos x}{x}\,\mathrm{d}x = \frac{1}{2}\int_1^\infty \frac{\sin 2x}{x}\,\mathrm{d}x$$

Step	Marks
Rewrites $\sin x \cos x = (1/2)\sin 2x$	1
Checks the Dirichlet conditions: $1/x$ is positive, decreasing, tends to 0, and the integral of $\sin 2x$ over $[1, t]$ is bounded (by 1)	1
Concludes convergence by Dirichlet's test	1

Integration by parts is also accepted: the boundary term $\cos 2x/(2x)$ tends to 0 and the remaining integral of $\cos 2x/x^2$ converges absolutely by comparison with $1/x^2$.

(b) The integral of $|\sin x \cos x|/x$ from 1 to ∞ diverges.

$$\int_{I_k} |\sin x \cos x|\,\mathrm{d}x = \frac{1}{2}, \quad I_k = \left[\frac{k\pi}{2}, \frac{(k+1)\pi}{2}\right]; \qquad \int_{I_k} \frac{|\sin x \cos x|}{x}\,\mathrm{d}x \geq \frac{2}{(k+1)\pi} \cdot \frac{1}{2} = \frac{1}{(k+1)\pi}$$

Step	Marks
Splits the range into intervals I_k of length $\pi/2$ and shows the integral of $\lvert\sin x \cos x\rvert$ over each I_k equals 1/2	1
Uses $x \leq (k+1)\pi/2$ on I_k to get a lower bound $1/((k+1)\pi)$ for each piece	1
Compares with the harmonic series and concludes divergence	1

An alternative route is to use $|\sin 2x| \geq \sin^2 2x = (1-\cos 4x)/2$, then show the integral of $1/(4x)$ diverges while the $\cos 4x/x$ part converges by Dirichlet's test. Award the same three marks if each step is justified.

A short closing remark that the integral in (a) is conditionally convergent earns no extra marks but shows good understanding.

Q5. Partial derivatives and continuity ($2+2=4$ marks)

(a) The partial derivatives at the origin are $f_x(0,0)=1$ and $f_y(0,0)=4$.

$$f(h,0)=\frac{h^2}{h}=h \Rightarrow f_x(0,0)=\lim_{h\to 0}\frac{h-0}{h}=1; \qquad f(0,h)=\frac{(2h)^2}{h}=4h \Rightarrow f_y(0,0)=\lim_{h\to 0}\frac{4h-0}{h}=4$$

Step	Marks
$f_x(0,0)=1$, computed from the limit definition using $f(h,0)=h$	1
$f_y(0,0)=4$, computed from the limit definition using $f(0,h)=4h$	1

Partial derivatives computed without using the definition (for example, keeping one variable fixed and differentiating with respect to the other) carry no marks, since the question asks for the definition.

(b) f is not continuous at $(0,0)$.

$$\text{Along } x=-y+y^2:\ x+y=y^2,\ \ x+2y=y+y^2, \quad f=\frac{(y+y^2)^2}{y^2}=(1+y)^2\to 1\neq 0=f(0,0)$$

Step	Marks
Chooses a path approaching $(0,0)$ on which $x+y$ tends to 0 but is not identically 0 (such as $x=-y+y^2$), and computes the limit 1 along it	1
States that the limit differs from $f(0,0)=0$, so f is not continuous	1

Any other valid path with a limit different from 0 (or two paths with different limits) earns full marks. Checking only the axes or lines $y=mx$ gives no credit for the first mark, since those limits do not show the discontinuity.

Q6. Temperature gradient ($2+2+2=6$ marks)

The gradient at $P(1,2,-1)$ is the common input to all three parts.

$$\nabla T = 5e^{-2x^2-y^2-2z^2}\langle -4x,-2y,-4z\rangle, \qquad \nabla T(P)=5e^{-8}\langle -4,-4,4\rangle = 20e^{-8}\langle -1,-1,1\rangle$$

(a) The rate of change toward $(1,1,0)$ is $20\sqrt{2}\,e^{-8}$ degrees per metre.

$$u=\frac{1}{\sqrt{2}}\langle 0,-1,1\rangle, \qquad D_uT(P)=20e^{-8}\cdot\frac{0+1+1}{\sqrt{2}}=20\sqrt{2}\,e^{-8}$$

(b) The temperature decreases most rapidly in the direction $(1,1,-1)/\sqrt{3}$, the unit vector along $-\nabla T(P)$.

(c) The maximum rate of decrease is $|\nabla T(P)| = 20\sqrt{3}\,e^{-8}$ degrees per metre.

Part	Step	Marks
(a)	Correct gradient of T and its value at P	1
(a)	Unit vector toward $(1,1,0)$ from P, dot product, answer $20\sqrt{2}\,e^{-8}$	1
(b)	Identifies the direction as $-\nabla T(P)$	1
(b)	Normalises to the unit vector $(1,1,-1)/\sqrt{3}$	1
(c)	States the maximum rate of decrease equals the magnitude of $\nabla T(P)$	1
(c)	Computes $20\sqrt{3}\,e^{-8}$	1

If the gradient in (a) is wrong but is used correctly in (b) and (c), award the method marks in (a) to (c) and withhold only the accuracy marks (follow-through). A direction in (b) that is not normalised loses its second mark. A sign change in the final answer cuts 1/2 mark. If the directions are wrong, no marks are awarded.

Q7. Gradient from directional derivatives (3 marks)

The gradient at (a,b,c) is $(2\sqrt{3},\ 6\sqrt{3},\ -4\sqrt{3})$.

$\nabla f = \langle p,q,r\rangle$: $\quad p+q+2r=0, \quad p-q-r=0, \quad p+q+r=4\sqrt{3} \implies p=2\sqrt{3},\ q=6\sqrt{3},\ r=-4\sqrt{3}$

Step	Marks
Writes $D_u f = \nabla f \cdot u$ for the three given unit vectors, with $\nabla f = (p,q,r)$	1
Clears denominators to get the correct linear system above	1
Solves the system and states $\nabla f = (2\sqrt{3}, 6\sqrt{3}, -4\sqrt{3})$	1

A quick solution: subtracting the first equation from the third gives $r=-4\sqrt{3}$; the second then gives $p=q+r$, and the third gives $q=6\sqrt{3}$. Check: $p+q+2r = 2\sqrt{3}+6\sqrt{3}-8\sqrt{3}=0$. In the final answer, if one of the coordinates is wrong, cut 1/2 mark. If there is a problem in the calculation of the coefficients (for example, forgetting to divide by $\sqrt{6}$ or $\sqrt{3}$ when forming the equations), only one mark is awarded.

Q8. Lagrange multipliers (4 marks)

The maximum of xy^2 on the circle is 2, at $(1,\sqrt{2})$ and $(1,-\sqrt{2})$; the minimum is -2, at $(-1,\sqrt{2})$ and $(-1,-\sqrt{2})$.

$$y^2 = 2\lambda x, \qquad 2xy = 2\lambda y, \qquad x^2+y^2=3$$

- Case $y=0$: $x=\pm\sqrt{3}$, and $f=0$ at both points.
- Case $y\neq 0$: the second equation gives $\lambda = x$, so $y^2=2x^2$. With $x^2+y^2=3$ this gives $x^2=1$ and $y^2=2$, so $f=xy^2=2x$, which equals 2 for $x=1$ and -2 for $x=-1$.

Step	Marks
Explains existence: f is continuous and the circle is closed and bounded (compact), so by the Extreme Value Theorem f attains a maximum and a minimum	1
Sets $\nabla f = \lambda \nabla g$ with $\nabla f = (y^2, 2xy)$ and $\nabla g = (2x, 2y)$, and writes the three equations	1
Solves by cases and finds all six critical points $(\pm\sqrt{3}, 0)$, $(\pm 1, \pm\sqrt{2})$	1
Compares values: maximum 2 and minimum -2, with the points where each occurs	1

Without the existence argument the student loses the first mark only. Missing the case $y = 0$ caps the solving mark at half, because the critical points $(\pm\sqrt{3}, 0)$ are never tested. Calculating the maximum without the minimum, or the minimum without the maximum, carries only 3 marks (1 mark deducted). 1/2 mark is deducted if any of the critical points is missing. In the final step, a calculation mistake costs 1 mark.

Mark summary

The eight questions add up to 40 marks.

Question	Topic	Mark split	Marks
Q1	Definitions	$1+1+1+1+1$	5
Q2	Riemann sums	$2+4$	6
Q3	Arc length and area	$3+3$	6
Q4	Improper integrals	$3+3$	6
Q5	Partial derivatives and continuity	$2+2$	4
Q6	Temperature gradient	$2+2+2$	6
Q7	Gradient from directional derivatives	3	3
Q8	Lagrange multipliers	4	4
Total			40